

JOHN MARIO MONTOYA ZAPATA

Master's student in Analytics, Specialist in Analytics, Civil Engineer

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PROFESSIONAL PROFILE

I am a Data Scientist passionate about AI and the deployment of ML models (MLOps), leveraging them to drive industry transformation. I specialize in developing innovative solutions that enhance processes and decision-making by integrating research with technology. I thrive in leadership roles within multidisciplinary teams and am committed to continuous learning to foster innovation and advance organizations.

WORK EXPERIENCE

Data Scientist with emphasis on AI (1 month)

TUYA S.A

Mar. 2025–present

Medellín, Ant

- Development and optimization of Machine Learning pipelines to automate and facilitate the integration of models in production, ensuring scalability, monitoring and continuous maintenance.
- Design and implementation of solutions based on generative AI to improve internal processes, automated content creation and personalization of financial products, focusing on innovation and efficiency.
- Support in the implementation of efficient, sustainable and scalable deployment architectures for Generative AI models and other ML algorithms.

Data scientist (16 months)

TUYA S.A (Risk analytics, Model risk)

Oct. 2023–Feb. 2025

Medellín, Ant

- Leadership in MLOps methodology and implementation of best practices for the development of AI models using tools such as MLflow, CI/CD, pyspark, DVC, Databricks, and agile methodologies (Scrum).
- Design, build and maintenance of the TuyaPy library in POO to facilitate the distribution and collaboration of the analytics team.
- Leadership in the design and development of Machine Learning models to estimate the probability of non-payment of customers in rotating and non-rotating products.

Data scientist (18 months)

COMERCIAL CARD S.A.S. - PTM

Mar. 2022–Sep 2023

Medellín, Ant

- Implementation of predictive models for portfolio forecasting, fraud detection, and time series analysis, with focus on systems such as SAGRILAFT and customer segmentation, including support in MLOps processes for production in Google Cloud.
- Participation in the creation of a Data Warehouse in BigQuery (GCP), designing ETL flows and using tools such as HEVO for its integration and feeding.

- Development of interactive reports in Power BI for financial KPIs, and automation of calculation processes in Python and KNIME through servers in Google Compute Engine.

Data analyst (20 months)

Early Warning System of Medellín and the Aburrá Valley - SIATA

Jul. 2020-Mar. 2022

Medellín, Ant

- Analysis of geospatial and temporal environmental information using dimensionality reduction and transformation techniques for decision making in early warning systems and risk management.
- Development of distributable packages in Python language using OOP, focused on agile data reading and analysis to support risk management and understanding of high frequency time series (20 Hz).

STUDIES

Master's student in engineering - Analytics

Universidad Nacional de Colombia

Ago. 2023-Dic. 2024

Medellín

Specialization in analytics

Universidad Nacional de Colombia

Ago. 2022-Jul. 2023

Medellín

Civil Engineering

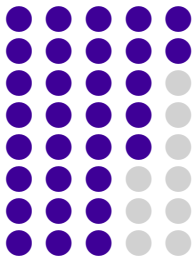
Universidad Nacional de Colombia

Feb. 2015-Nov. 2020

Medellín

TECH SKILLS

Python
SQL
Power BI
KNIME
Cloud computing
Pyspark
RAG
HuggingFace



AREAS OF INTEREST

Machine Learning Generative IA MLOps
Data Science Data Engineering Statistics
Mathematical modeling Mathematical modeling
Analytics Programming Cloud Computing

COURSES

Academy Accreditation - Databricks Lakehouse Fundamentals

Databricks

📅 Mar. 2024 📍 databricks.com

Detección de objetos

Coursera

📅 Oct. 2023 📍 coursera.org

Ruta Científico de datos

Platzi

📅 Jul. 2022-Presente 📍 platzi.com

Misión TIC 2022

MINTIC

📅 Dic. 2021 📍 misiontic2022.gov.co

Data Science for All (DS4A)

MINTIC & Correlation One

📅 Sept. 2021 📍 correlation-one.com

Applied Machine Learning in Python

University of Michigan

📅 Sept. 2020 📍 coursera.org

Applied Plotting, Charting and Data Representation in Python

University of Michigan

📅 Jul 2020 📍 coursera.org

LANGUAGES

Spanish: Native English: B1-Intermediate

ACADEMIC EXPERIENCE

Spatio-temporal characterization of turbulent flows in the Aburrá Valley.

Undergraduate thesis

📅 Oct. 2019–Abr. 2020 (7 months) 📍 Medellín

- Statistical analysis of data collected by sonic anemometers of high temporal resolution (20 Hz).
- Application of frequency analysis techniques (Fourier transform, principal component analysis) in environmental time series.
- Implementation of neural networks for modeling turbulent flow time series.

AWARDS

Award “Best Undergraduate Degree Works” - Civil Engineering Program (Medellin Campus), version XXX

National University of Colombia

📅 Apr. 2021 📍 Medellín

“Michel Hermelin Arbaux” Award, version 2021

Colombian Academy of Exact, Physical and Natural Sciences

📅 Nov. 2021 📍 Medellín

REFERENCES

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